(a)

$$H/h = 5.200\, b^\dagger b - \frac{0.210}{2} b^\dagger b^\dagger b b + 7.100\, a^\dagger a + 0.090\,(a^\dagger b + a b^\dagger)$$

transmon: junction and shunt capacitor

readout resonator

rotating-wave form

coupling capacitor: $g \propto C_g/\sqrt{C_\Sigma C_r}$

(b)

one-excitation block (GHz)

	q	r
q	5.200	0.090
r	0.090	7.100

mode	bare (GHz)	dressed (GHz)
q	5.200	5.196
r	7.100	7.104

(c)

$$H_d(t)/h = \Omega(t)\cos(2\pi f_d t)\,(b + b^\dagger)$$

$$\Omega(t) \propto C_d\, V_d(t)\, \langle 0|n|1\rangle$$

assumed: T_1, T_2^*, readout fidelity